

TPR Drop Analysis After New Scanner Introduction

Objective

Understand why slice-wise TPR drops sharply for one site after a new scanner is introduced, while service metrics remain normal.

Learning Goals

- * Distinguish data/model issues from service issues
- * Master 4 diagnostic approaches
- * Implement 3 mitigation strategies

The Problem

Slice-wise TPR drops sharply for one site after a new scanner is introduced

Key Facts

- * Before: Site 3 TPR = 0.92
- * After: Site 3 TPR = 0.65 (DEGRADED)
- * Service Latency: Normal (50ms)
- * Error Rate: Normal (0.1%)

The Challenge

System appears healthy, but predictions are wrong for Site 3!

Real-World Use Case

Medical Imaging Scenario

- * 3 hospital sites with different scanners
- * ML model for disease detection
- * New scanner deployed at Site 3

The Risk

- * False Negatives = Missed disease
- * Patients at Site 3 receiving worse care
- * System appears healthy but predictions are wrong

Industry Relevance

Healthcare, Radiology, Pathology, and any ML system with diverse data sources

Input Data / Inputs

Multi-Site Scanner Data

Scanner	Site	Mean	Std	Type
Scanner_A	Site_1	100	15	Original
Scanner_B	Site_2	100	15	Original
Scanner_C	Site_3	120	25	NEW

Key Difference

New scanner has DIFFERENT distribution (Mean: 120 vs 100, Std: 25 vs 15)

Concepts Used (High Level)

Concept	Purpose
TPR (Recall)	Measure positive class detection rate
Covariate Shift	Input distribution change
Slice-based Monitoring	Per-subgroup evaluation
KL Divergence	Quantify distribution difference
Safe Fallback	Human-in-the-loop protection
Domain Adaptation	Model robustness improvement

Concepts Breakdown (Simple)

TPR = True Positive Rate

"Out of 100 disease cases, how many did we correctly detect?"

Covariate Shift

"Training data looks different from production data"

Slice-based Monitoring

"Check accuracy for each group separately, not just overall"

Safe Fallback

"When unsure, let a human expert decide"

Step-by-Step Solution Flow

1. OBSERVE

- * TPR drops for Site 3
- * Service metrics remain normal

2. DIAGNOSE

- * Feature distribution comparison
- * Confidence calibration check
- * Confusion matrix breakdown
- * Temporal trend analysis

3. MITIGATE

- * P0: Safe fallback (human review)
- * P1: Preprocessing normalization
- * P2: Domain adaptation / retraining

Code Logic Summary

Main Sections

- * 1. Simulate Data: 3 scanners with different distributions
- * 2. Train Model: Only on original scanners (A, B)
- * 3. Evaluate per Slice: TPR for each scanner
- * 4. Check Service Metrics: Confirm infrastructure healthy
- * 5. Run Diagnostics: 4 diagnostic functions
- * 6. Apply Mitigations: 3 mitigation strategies

Key Functions

`simulate_scanner_data()`, `train_model_on_original_scanners()`, `evaluate_per_scanner()`, `diagnostic_*`(), `mitigation_*`()

Important Functions & Parameters

evaluate_per_scanner(df, model, scaler)

Calculates TPR per scanner slice, returns confusion matrix breakdown

diagnostic_feature_distribution(df)

Computes KL divergence per feature. Threshold > 0.1 = significant shift

mitigation_safe_fallback(df, model, scaler, confidence_threshold=0.7)

Routes low-confidence predictions to human review. Default threshold: 0.7

Execution Output

TPR per Scanner

Scanner	TPR	Status
Scanner_A	0.91	Normal
Scanner_B	0.90	Normal
Scanner_C_NEW	0.65	DEGRADED

Diagnostic: KL Divergence

feature_0: 0.55 (significant shift detected!)

feature_1: 1.21 (HIGH distribution shift!)

Observations & Insights

1. Overall Accuracy Hides Subgroup Issues

If we only monitored overall accuracy, we would NOT have detected this issue

2. Normal Service != Healthy Model

System appears healthy but predictions are wrong (silent failure mode)

3. New Scanner = Distribution Shift

KL Divergence confirms the new scanner data is very different

4. Model is Confidently Wrong

High confidence on false negatives = miscalibration

Advantages & Limitations

Advantages

- * Detects silent failures
- * Quantifies distribution shift
- * Provides actionable mitigations
- * Enables safe deployment

Limitations

- * Requires labeled data for adaptation
- * KL divergence sensitive to binning
- * Human review does not scale infinitely
- * Preprocessing may not capture all differences

Interview Key Takeaways

Q1: Why is this a data/model issue, not service issue?

Service metrics are normal. TPR drop is isolated to one site/scanner, pointing to data distribution shift.

Q2: What diagnostics would you run?

KL divergence, confidence calibration, per-scanner confusion matrices, temporal trend analysis.

Q3: What is your first mitigation?

Safe fallback - route low-confidence predictions to human review for immediate protection.

Conclusion

Root Cause

Covariate shift from new scanner with different data characteristics

Evidence

- * Service metrics normal - Infrastructure healthy
- * TPR drop isolated to Site 3 - Not global failure
- * High KL divergence - Distribution shift confirmed
- * Temporal correlation - New scanner is cause

Action Plan

- * P0: Safe fallback (human review)
- * P1: Preprocessing normalization
- * P2: Domain adaptation / retraining

Key Takeaway

Always monitor slice-wise metrics. Service health $\neq$ model health. Implement safe fallbacks for high-stakes applications.